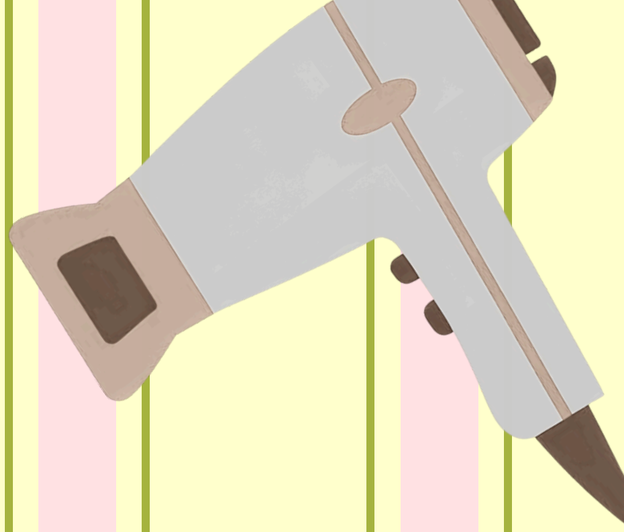
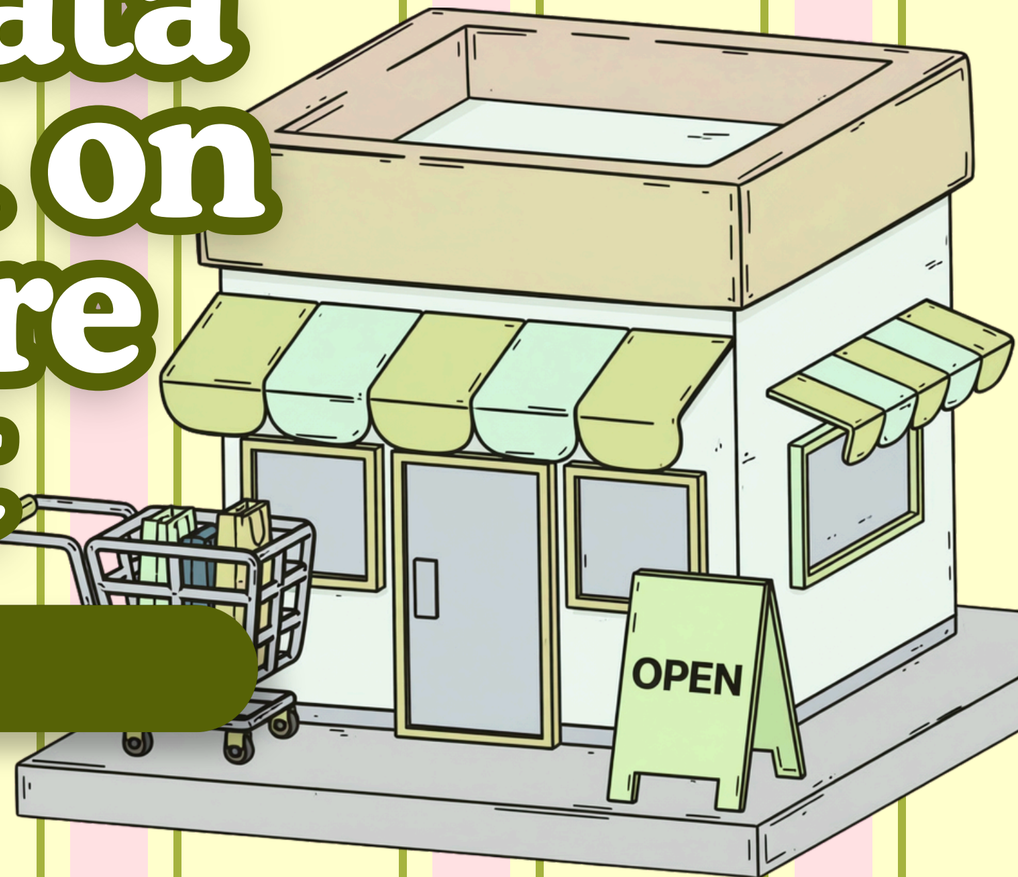
MDA3083 Predective Analytics

Data Reduction and Big Data Using PCA on Superstore Dataset



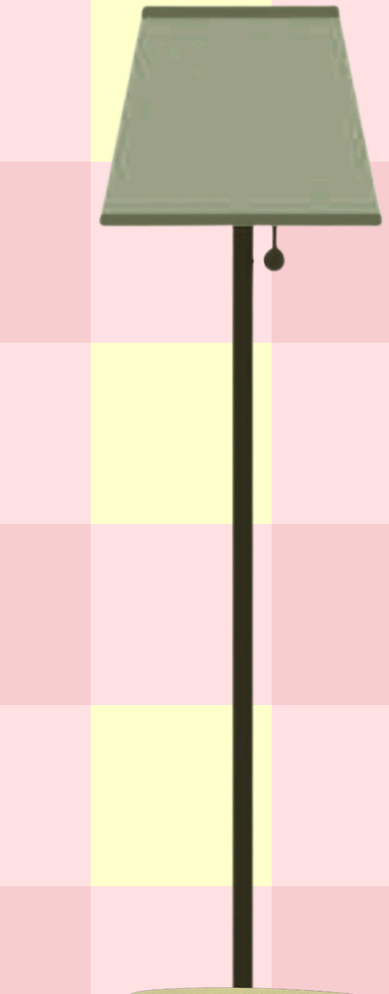


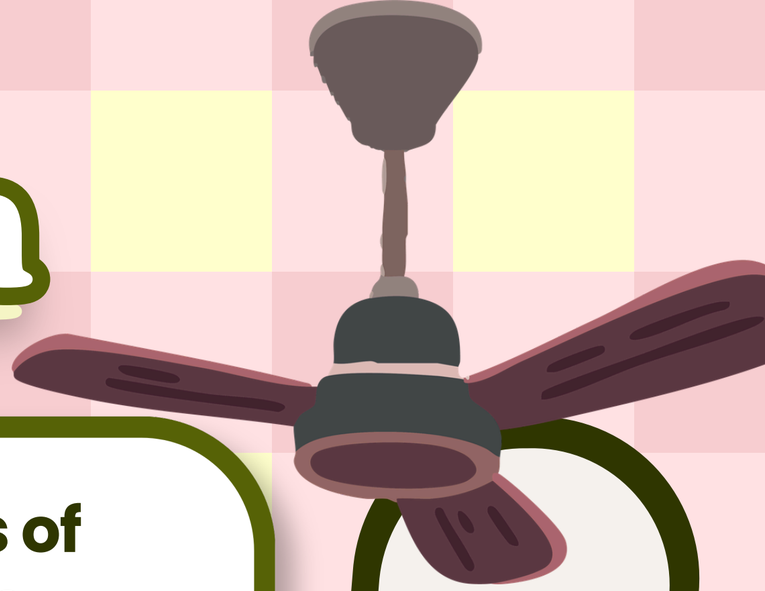


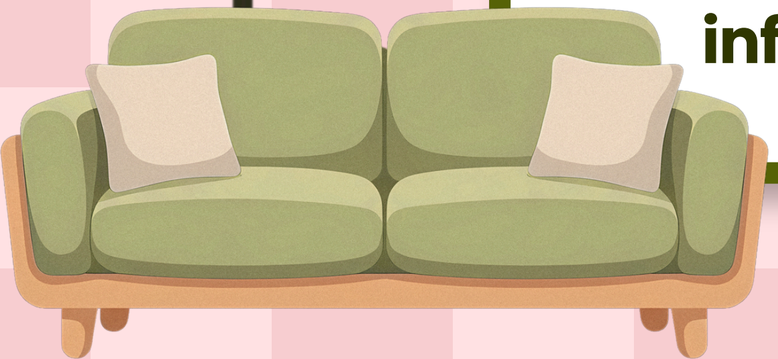


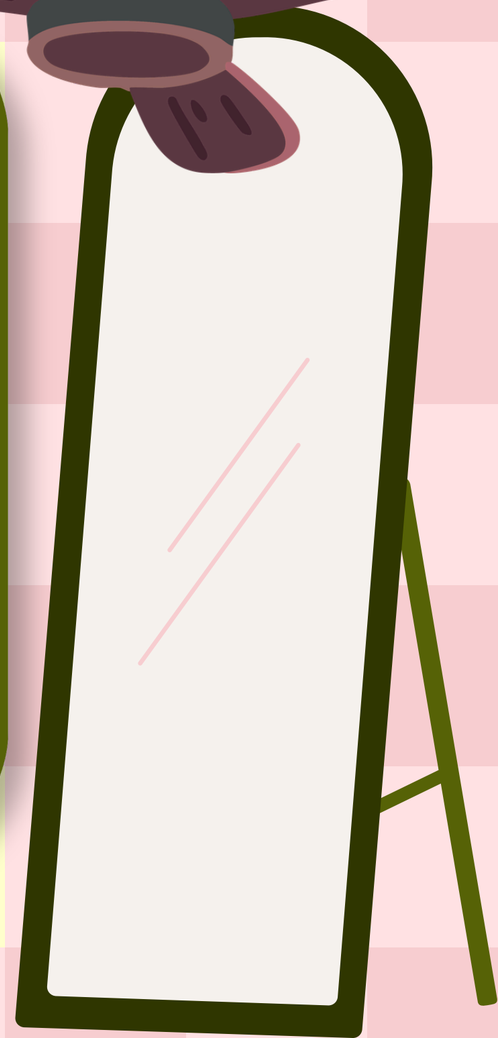
Group 7

Introduction







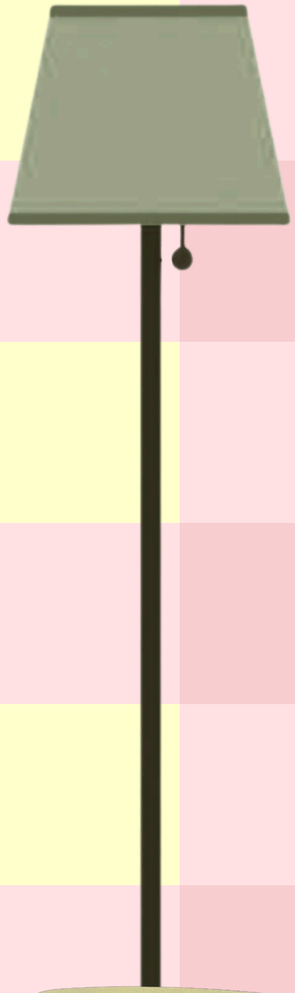


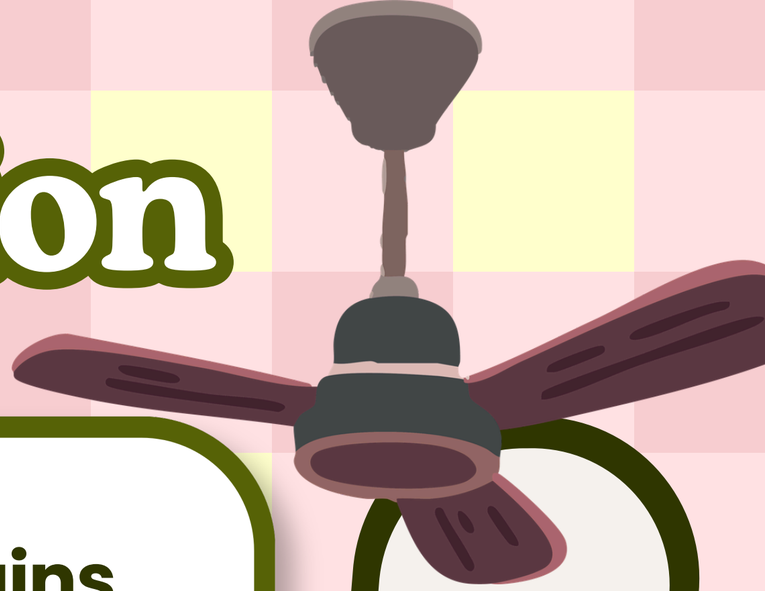
Businesses generate large amounts of data that can be difficult to analyze. Principle Component Analysis (PCA) helps reduce the number of variables while keeping the important information. In this study, PCA was applied to the Superstore Dataset to reduce the number of variables and identify the main factors that influence retail business performance.

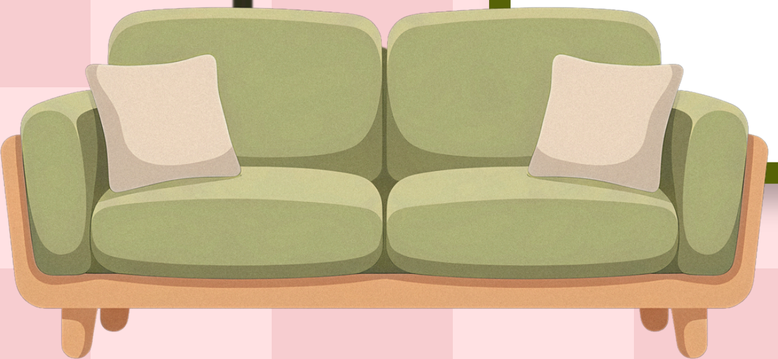
Dataset Information

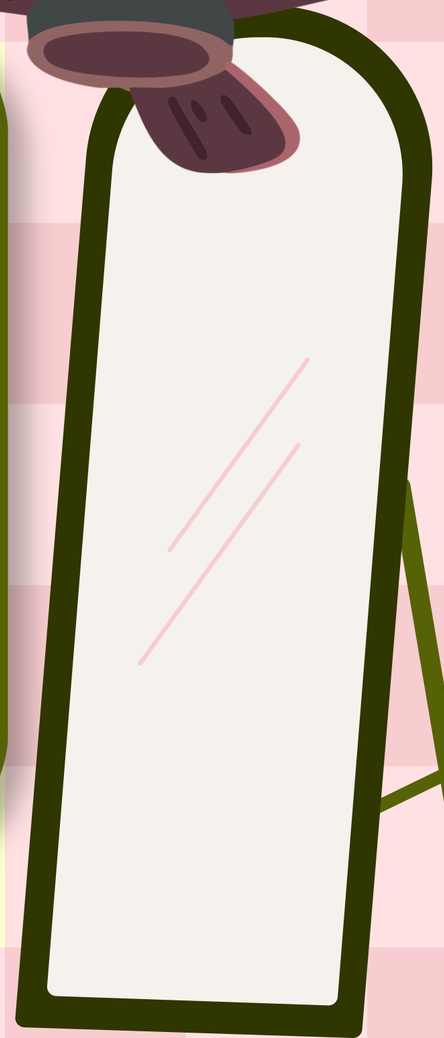
Item	Description
Dataset	Superstore Dataset
Source	Kaggle
Industry	Retail Business
Observations	9,994
Variables	21
Software	R 4.6.0

Data Description









The Superstore Dataset contains 9,994 retail sales transactions from a superstore in the United States between 2014 until 2017. It includes customer, product, order, and financial information.

Methodology

1. Dataset and Data Source
2. Data Exploration
3. Prinncipal Component Analysis (PCA)
4. Eigenvalue Analysis
5. Data Visualisation
6. Component Loadings



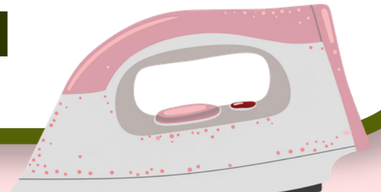
Data Exploration



- Imported dataset into R
- Conducted Exploratory Data Analysis (EDA)
- Checking for missing values and duplicate records
- Result:
  - No missing values found
  - No duplicate observations
  - No data cleaning required

```
> # Total missing values
> sum(is.na(superstore))
[1] 0
```

```
> # Duplicate Records
> sum(duplicated(superstore))
[1] 0
```



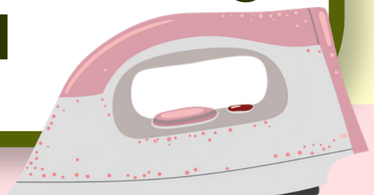
Examined:

- Variable types
- Number of observations
- Distribution of numerical variables
- Descriptive statistics (minimum, maximum, mean and median)

Data Exploration



- Imported dataset into R
- Conducted EDA
- Checking for missing values and duplicate records
- Result:
  - No missing values found
  - No duplicate observations
  - No data cleaning required



```
> # Variable Types
> str(superstore)
tibble [9,994 × 21] (S3: tbl_df/tbl/data.frame)
 $ Row ID       : num [1:9994] 1 2 3 4 5 6 7 8 9 10 ...
 $ Order ID     : chr [1:9994] "CA-2016-152156" "CA-2016-152156" "CA-2016-138688" "US-2015-108966
" ...
 $ Order Date   : POSIXct[1:9994], format: "2016-11-08" "2016-11-08" ...
 $ Ship Date    : POSIXct[1:9994], format: "2016-11-11" "2016-11-11" ...
 $ Ship Mode    : chr [1:9994] "Second Class" "Second Class" "Second Class" "Standard Class" ...
 $ Customer ID  : chr [1:9994] "CG-12520" "CG-12520" "DV-13045" "SO-20335" ...
 $ Customer Name: chr [1:9994] "Claire Gute" "Claire Gute" "Darrin Van Huff" "Sean O'Donnell" ...
 $ Segment      : chr [1:9994] "Consumer" "Consumer" "Corporate" "Consumer" ...
 $ Country      : chr [1:9994] "United States" "United States" "United States" "United States" ..
.
 $ City         : chr [1:9994] "Henderson" "Henderson" "Los Angeles" "Fort Lauderdale" ...
 $ State        : chr [1:9994] "Kentucky" "Kentucky" "California" "Florida" ...
 $ Postal Code  : num [1:9994] 42420 42420 90036 33311 33311 ...
 $ Region       : chr [1:9994] "South" "South" "West" "South" ...
 $ Product ID   : chr [1:9994] "FUR-BO-10001798" "FUR-CH-10000454" "OFF-LA-10000240" "FUR-TA-1000
0577" ...
 $ Category     : chr [1:9994] "Furniture" "Furniture" "Office Supplies" "Furniture" ...
 $ Sub-Category : chr [1:9994] "Bookcases" "Chairs" "Labels" "Tables" ...
 $ Product Name : chr [1:9994] "Bush Somerset Collection Bookcase" "Hon Deluxe Fabric Upholstered
 Stacking Chairs, Rounded Back" "Self-Adhesive Address Labels for Typewriters by Universal" "Bret
ford CR4500 Series Slim Rectangular Table" ...
 $ Sales        : num [1:9994] 262 731.9 14.6 957.6 22.4 ...
 $ Quantity     : num [1:9994] 2 3 2 5 2 7 4 6 3 5 ...
 $ Discount     : num [1:9994] 0 0 0 0.45 0.2 0 0 0.2 0.2 0 ...
 $ Profit       : num [1:9994] 41.91 219.58 6.87 -383.03 2.52 ...
```

```
> # Summary Statistics
> summary(superstore)
     Row ID          Order ID          Order Date
 Min.   :   1   Length   :9994   Min.   :2014-01-03 00:00:00
 1st Qu.:2499   N.unique :5009   1st Qu.:2015-05-23 00:00:00
 Median :4998   N.blank  :   0   Median :2016-06-26 00:00:00
 Mean   :4998   Min.nchar:  14   Mean   :2016-04-29 23:41:50
 3rd Qu.:7496   Max.nchar:  14   3rd Qu.:2017-05-14 00:00:00
 Max.   :9994                    Max.   :2017-12-30 00:00:00
   Ship Date                     Ship Mode        Customer ID
 Min.   :2014-01-07 00:00:00   Length   :9994   Length   :9994
 1st Qu.:2015-05-27 00:00:00   N.unique :   4   N.unique : 793
 Median :2016-06-29 00:00:00   N.blank  :   0   N.blank  :   0
 Mean   :2016-05-03 23:06:58   Min.nchar:   8   Min.nchar:   8
 3rd Qu.:2017-05-18 00:00:00   Max.nchar:  14   Max.nchar:   9
 Max.   :2018-01-05 00:00:00
   Customer Name       Segment           Country             City
 Length   :9994   Length   :9994   Length   :9994   Length   :9994
 N.unique : 793   N.unique :   3   N.unique :   1   N.unique : 531
 N.blank  :   0   N.blank  :   0   N.blank  :   0   N.blank  :   0
 Min.nchar:   7   Min.nchar:   8   Min.nchar:  13   Min.nchar:   4
 Max.nchar:  22   Max.nchar:  11   Max.nchar:  13   Max.nchar:  17

      State          Postal Code         Region          Product ID
 Length   :9994   Min.   : 1040   Length   :9994   Length   :9994
 N.unique :  49   1st Qu.:23223   N.unique :   4   N.unique :1862
 N.blank  :   0   Median :56431   N.blank  :   0   N.blank  :   0
 Min.nchar:   4   Mean   :55190   Min.nchar:   4   Min.nchar:  15
 Max.nchar:  20   3rd Qu.:90008   Max.nchar:   7   Max.nchar:  15
                  Max.   :99301
     Category       Sub-Category       Product Name         Sales
 Length   :9994   Length   :9994   Length   :9994   Min.   :    0.444
 N.unique :   3   N.unique :  17   N.unique :1850   1st Qu.:   17.280
 N.blank  :   0   N.blank  :   0   N.blank  :   0   Median :   54.490
 Min.nchar:   9   Min.nchar:   3   Min.nchar:   5   Mean   :  229.858
 Max.nchar:  15   Max.nchar:  11   Max.nchar: 127   3rd Qu.:  209.940
                                                    Max.   :22638.480
    Quantity        Discount          Profit
 Min.   : 1.00   Min.   :0.0000   Min.   :-6599.978
 1st Qu.: 2.00   1st Qu.:0.0000   1st Qu.:    1.729
 Median : 3.00   Median :0.2000   Median :    8.666
 Mean   : 3.79   Mean   :0.1562   Mean   :   28.657
 3rd Qu.: 5.00   3rd Qu.:0.2000   3rd Qu.:   29.364
 Max.   :14.00   Max.   :0.8000   Max.   : 8399.976
```

Examined:

- Variable types
- Number of observations
- Descriptive statistics (minimum, maximum, mean and median)

```
> dim(superstore)
[1] 9994   21
```

PCA & Eigenvalues Analysis

Principal Component	Eigenvalue	Variance (%)	Cumulative (%)
PC1	1.590	39.75	39.75
PC2	1.059	26.47	66.22
PC3	0.881	22.03	88.25
PC4	0.470	11.75	100

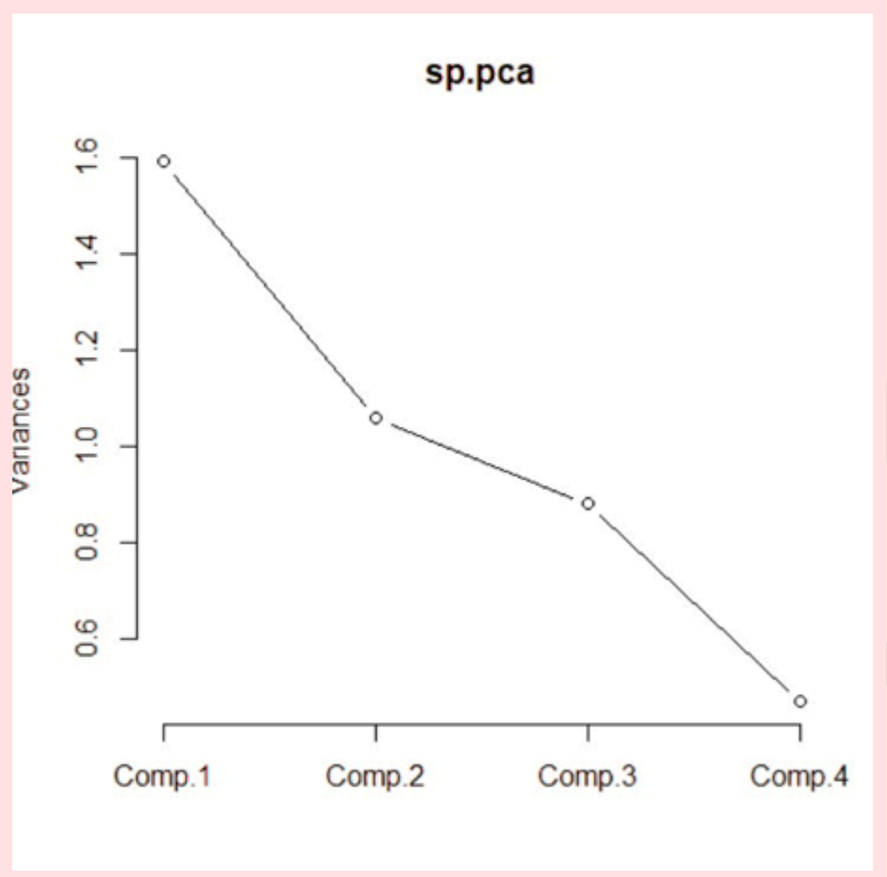
Selected Components

- **PC1 = 1.590**
- **PC2 = 1.059**

Result

- **Reduced 4 component → 3 PC**
- **Retained 88.25% of dataset information**

Scree Plot



Component Loadings



Variable	PC1	PC2
Sales	0.643	0.218
Quantity	0.289	0.631
Discount	-0.27	0.719
Profit	0.655	-0.195

Interpretation

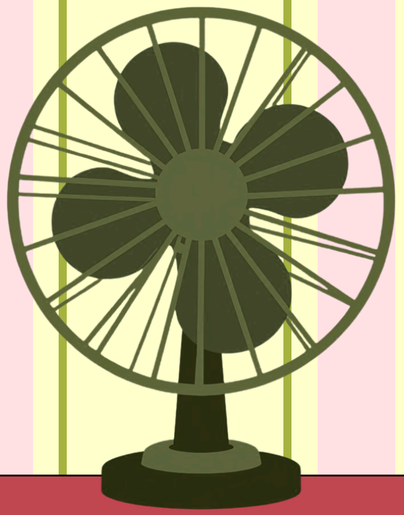
PC1 – Business Performance Component

- **Dominated by Sales and Profit.**
- **Represents revenue generation and overall business performance.**

PC2 – Discount & Purchasing Behaviour

- **Dominated by Discount and Quantity.**
- **Represents the effect of promotions on customer purchasing patterns.**

Correlation Analysis

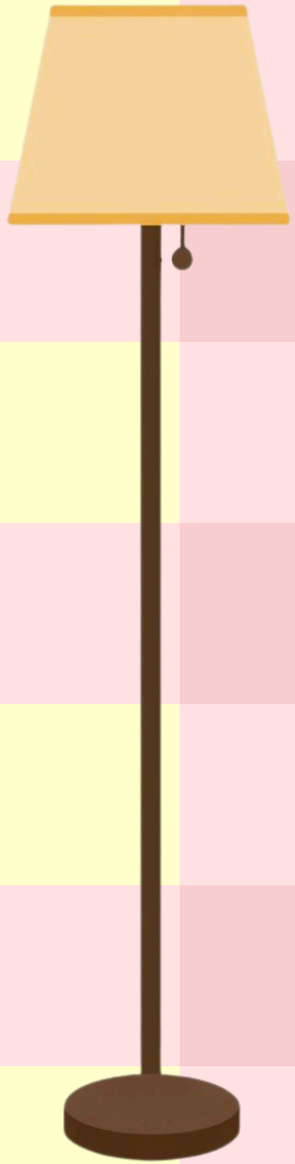


Component	Correlation
Sales + PC1	0.81 (strong positive correlation)
Sales + PC2	0.22 (weak positive correlation)

- **Sales+PC1 : Indicates that PC1 represents overall business performance (Sales and Profit).**
- **Sales+PC2 : Suggests that PC2 reflects discount strategies and purchasing behavior rather than sales performance.**

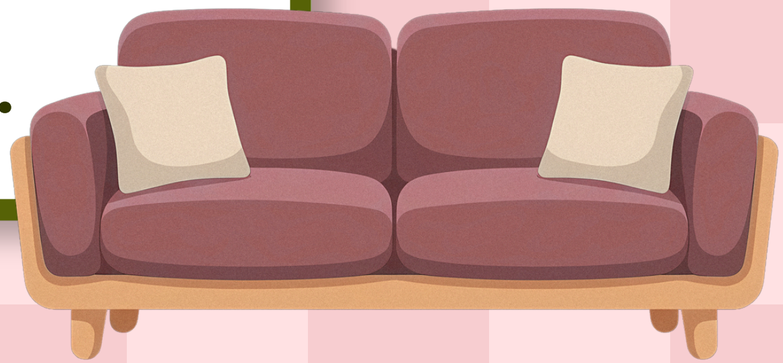


Business Implications



Business Insights

- Sales and Profit are strongly related.
- Discount influences purchasing quantity.
- PCA reduced four variables into two components.
- Simplifies business analysis.
- Supports better decision-making.



Conclusion



- PCA successfully reduced the dataset from 4 variables to 2 principal components.
- Two components retained 66.22% of the original information.

- PC1: Business Performance (Sales & Profit)
- PC2: Discount & Purchasing Behaviour
- PCA effectively simplifies retail business performance analysis.



Thank You!

From us, Group 7 :)